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An integrated encyclopedia of DNA elements in the human genome

The ENCODE Project Consortium*

The human genome encodes the blueprint of life, but the function of the vast majority of its nearly three billion bases is unknown. The Encyclopedia of DNA Elements (ENCODE) project has systematically mapped regions of transcription, transcription factor association, chromatin structure and histone modification. These data enabled us to assign biochemical functions for 80% of the genome, in particular outside of the well-studied protein-coding regions. Many discovered candidate regulatory elements are physically associated with one another and with expressed genes, providing new insights into the mechanisms of gene regulation. The newly identified elements also show a statistical correspondence to sequence variants linked to human disease, and can thereby guide interpretation of this variation. Overall, the project provides new insights into the organization and regulation of our genes and genome, and is an expansive resource of functional annotations for biomedical research.

The human genome sequence provides the underlying code for human biology. Despite intensive study, especially in identifying protein-coding genes, our understanding of the genome is far from complete, particularly with regard to non-coding RNAs, alternatively spliced transcripts and regulatory sequences. Systematic analyses of transcripts and regulatory information are essential for the identification of genes and regulatory regions, and are an important resource for the study of human biology and disease. Such analyses can also provide comprehensive views of the organization and variability of genes and regulatory information across cellular contexts, species and individuals.

ENCODE
Encyclopedia of DNA Elements
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The Encyclopedia of DNA Elements (ENCODE) project aims to delineate all functional elements encoded in the human genome[1–3]. Operationally, we define a functional element as a discrete genome segment that encodes a defined product (for example, protein or non-coding RNA) or displays a reproducible biochemical signature (for example, protein binding, or a specific chromatin structure). Comparative genomic studies suggest that 3–8% of bases are under purifying (negative) selection[4–8] and therefore may be functional, although other analyses have suggested much higher estimates[9–11]. In a pilot phase covering 1% of the genome, the ENCODE project annotated 60% of mammalian evolutionarily constrained bases, but also identified many additional putative functional elements without evidence of constraint[2]. The advent of more powerful DNA sequencing technologies now enables whole-genome and more precise analyses with a broad repertoire of functional assays.

Here we describe the production and initial analysis of 1,640 data sets designed to annotate functional elements in the entire human genome. We integrate results from diverse experiments within cell types, related experiments involving 147 different cell types, and all ENCODE data with other resources, such as candidate regions from genome-wide association studies (GWAS) and evolutionarily constrained regions. Together, these efforts reveal important features about the organization and function of the human genome, summarized below.

- The vast majority (80.4%) of the human genome participates in at least one biochemical RNA- and/or chromatin-associated event in at least one cell type. Much of the genome lies close to a regulatory event: 95% of the genome lies within 8 kilobases (kb) of a DNA–protein interaction (as assayed by bound ChIP-seq motifs or DNase I footprints), and 99% is within 1.7 kb of at least one of the biochemical events measured by ENCODE.
- Primate-specific elements as well as elements without detectable mammalian constraint show, in aggregate, evidence of negative selection; thus, some of them are expected to be functional.
- Classifying the genome into seven chromatin states indicates an initial set of 399,124 regions with enhancer-like features and 70,292 regions with promoter-like features, as well as hundreds of thousands of quiescent regions. High-resolution analyses further subdivide the genome into thousands of narrow states with distinct functional properties.
- It is possible to correlate quantitatively RNA sequence production and processing with both chromatin marks and transcription factor binding at promoters, indicating that promoter functionality can explain most of the variation in RNA expression.
- Many non-coding variants in individual genome sequences lie in ENCODE-annotated functional regions; this number is at least as large as those that lie in protein-coding genes.
- Single nucleotide polymorphisms (SNPs) associated with disease by GWAS are enriched within non-coding functional elements, with a majority residing in or near ENCODE-defined regions that are outside of protein-coding genes. In many cases, the disease phenotypes can be associated with a specific cell type or transcription factor.

ENCODE data production and initial analyses

Since 2007, ENCODE has developed methods and performed a large number of sequence-based studies to map functional elements across the human genome[3]. The elements mapped (and approaches used) include RNA transcribed regions (RNA-seq, CAGE, RNA-PET and manual annotation), protein-coding regions (mass spectrometry), transcription-factor-binding sites (ChIP-seq and DNase-seq), chromatin structure (DNase-seq, FAIRE-seq, histone ChIP-seq and MNase-seq), and DNA methylation sites (RRBS assay) (Box 1 lists methods and abbreviations; Supplementary Table 1, section P, details production statistics)[3]. To compare and integrate results across the different laboratories, data production efforts focused on two selected

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BOX 1

ENCODE abbreviations

RNA-seq. Isolation of RNA sequences, often with different purification techniques to isolate different fractions of RNA followed by high-throughput sequencing.

CAGE. Capture of the methylated cap at the 5′ end of RNA, followed by high-throughput sequencing of a small tag adjacent to the 5′ methylated caps. 5′ methylated caps are formed at the initiation of transcription, although other mechanisms also methylate 5′ ends of RNA.

RNA-PET. Simultaneous capture of RNAs with both a 5′ methyl cap and a poly(A) tail, which is indicative of a full-length RNA. This is then followed by sequencing a short tag from each end by high-throughput sequencing.

ChIP-seq. Chromatin immunoprecipitation followed by sequencing. Specific regions of crosslinked chromatin, which is genomic DNA in complex with its bound proteins, are selected by using an antibody to a specific epitope. The enriched sample is then subjected to high-throughput sequencing to determine the regions in the genome most often bound by the protein to which the antibody was directed. Most often used are antibodies to any chromatin-associated epitope, including transcription factors, chromatin binding proteins and specific chemical modifications on histone proteins.

DNase-seq. Adaption of established regulatory sequence assay to modern techniques. The DNase I enzyme will preferentially cut live chromatin preparations at sites where nearby there are specific (non-histone) proteins. The resulting cut points are then sequenced using high-throughput sequencing to determine those sites 'hypersensitive' to DNase I, corresponding to open chromatin.

FAIRE-seq. Formaldehyde assisted isolation of regulatory elements. FAIRE isolates nucleosome-depleted genomic regions by exploiting the difference in crosslinking efficiency between nucleosomes (high) and sequence-specific regulatory factors (low). FAIRE consists of crosslinking, phenol extraction, and sequencing the DNA fragments in the aqueous phase.

RRBS. Reduced representation bisulphite sequencing. Bisulphite treatment of DNA sequence converts unmethylated cytosines to uracil. To focus the assay and save costs, specific restriction enzymes that cut around CpG dinucleotides can reduce the genome to a portion specifically enriched in CpGs. This enriched sample is then sequenced to determine the methylation status of individual cytosines quantitatively.

Tier 1. Tier 1 cell types were the highest-priority set and comprised three widely studied cell lines: K562 erythroleukaemia cells; GM12878, a B-lymphoblastoid cell line that is also part of the 1000 Genomes project (http://1000genomes.org)[55]; and the H1 embryonic stem cell (H1 hESC) line.

Tier 2. The second-priority set of cell types in the ENCODE project which included HeLa-S3 cervical carcinoma cells, HepG2 hepatoblastoma cells and primary (non-transformed) human umbilical vein endothelial cells (HUVECs).

Tier 3. Any other ENCODE cell types not in tier 1 or tier 2.

sets of cell lines, designated 'tier 1' and 'tier 2' (Box 1). To capture a broader spectrum of biological diversity, selected assays were also executed on a third tier comprising more than 100 cell types including primary cells. All data and protocol descriptions are available at http://www.encodeproject.org/, and a User's Guide including details of cell-type choice and limitations was published recently[3].

Integration methodology

For consistency, data were generated and processed using standardized guidelines, and for some assays, new quality-control measures were designed (see refs 3, 12 and http://encodeproject.org/ENCODE/dataStandards.html; A. Kundaje, personal communication). Uniform data-processing methods were developed for each assay (see Supplementary Information; A. Kundaje, personal communication), and most assay results can be represented both as signal information (a per-base estimate across the genome) and as discrete elements (regions computationally identified as enriched for signal). Extensive processing pipelines were developed to generate each representation (M. M. Hoffman *et al.*, manuscript in preparation and A. Kundaje, personal communication). In addition, we developed the irreproducible discovery rate (IDR)[13] measure to provide a robust and conservative estimate of the threshold where two ranked lists of results from biological replicates no longer agree (that is, are irreproducible), and we applied this to defining sets of discrete elements. We identified, and excluded from most analyses, regions yielding untrustworthy signals likely to be artefactual (for example, multicopy regions). Together, these regions comprise 0.39% of the genome (see Supplementary Information). The poster accompanying this issue represents different ENCODE-identified elements and their genome coverage.

Transcribed and protein-coding regions

We used manual and automated annotation to produce a comprehensive catalogue of human protein-coding and non-coding RNAs as well as pseudogenes, referred to as the GENCODE reference gene set[14,15] (Supplementary Table 1, section U). This includes 20,687 protein-coding genes (GENCODE annotation, v7) with, on average, 6.3 alternatively spliced transcripts (3.9 different protein-coding transcripts) per locus. In total, GENCODE-annotated exons of protein-coding genes cover 2.94% of the genome or 1.22% for protein-coding exons. Protein-coding genes span 33.45% from the outermost start to stop codons, or 39.54% from promoter to poly(A) site. Analysis of mass spectrometry data from K562 and GM12878 cell lines yielded 57 confidently identified unique peptide sequences in intergenic regions relative to GENCODE annotation. Taken together with evidence of pervasive genome transcription[16], these data indicate that additional protein-coding genes remain to be found.

In addition, we annotated 8,801 automatically derived small RNAs and 9,640 manually curated long non-coding RNA (lncRNA) loci[17]. Comparing lncRNAs to other ENCODE data indicates that lncRNAs are generated through a pathway similar to that for protein-coding genes[17]. The GENCODE project also annotated 11,224 pseudogenes, of which 863 were transcribed and associated with active chromatin[18].

RNA

We sequenced RNA[16] from different cell lines and multiple subcellular fractions to develop an extensive RNA expression catalogue. Using a conservative threshold to identify regions of RNA activity, 62% of genomic bases are reproducibly represented in sequenced long (>200 nucleotides) RNA molecules or GENCODE exons. Of these bases, only 5.5% are explained by GENCODE exons. Most transcribed bases are within or overlapping annotated gene boundaries (that is, intronic), and only 31% of bases in sequenced transcripts were intergenic[16].

We used CAGE-seq (5′ cap-targeted RNA isolation and sequencing) to identify 62,403 transcription start sites (TSSs) at high confidence (IDR of 0.01) in tier 1 and 2 cell types. Of these, 27,362 (44%) are within 100 base pairs (bp) of the 5′ end of a GENCODE-annotated transcript or previously reported full-length messenger RNA. The remaining regions predominantly lie across exons and 3′ untranslated regions (UTRs), and some exhibit cell-type-restricted expression; these may represent the start sites of novel, cell-type-specific transcripts.

Finally, we saw a significant proportion of coding and non-coding transcripts processed into steady-state stable RNAs shorter than 200 nucleotides. These precursors include transfer RNA, microRNA, small nuclear RNA and small nucleolar RNA (tRNA, miRNA, snRNA and snoRNA, respectively) and the 5′ termini of these processed products align with the capped 5′ end tags[16].

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Table 1 | Summary of transcription factor classes analysed in ENCODE

Acronym	Description	Factors analysed
ChromRem	ATP-dependent chromatin complexes	5
DNARep	DNA repair	3
HISase	Histone acetylation, deacetylation or methylation complexes	8
Other	Cyclin kinase associated with transcription	1
Pol2	Pol II subunit	1 (2 forms)
Pol3	Pol III-associated	6
TFNS	General Pol II-associated factor, not site-specific	8
TFSS	Pol II transcription factor with sequence-specific DNA binding	87

Protein bound regions

To identify regulatory regions directly, we mapped the binding locations of 119 different DNA-binding proteins and a number of RNA polymerase components in 72 cell types using ChIP-seq (Table 1, Supplementary Table 1, section N, and ref. 19); 87 (73%) were sequence-specific transcription factors. Overall, 636,336 binding regions covering 231 megabases (Mb; 8.1%) of the genome are enriched for regions bound by DNA-binding proteins across all cell types. We assessed each protein-binding site for enrichment of known DNA-binding motifs and the presence of novel motifs. Overall, 86% of the DNA segments occupied by sequence-specific transcription factors contained a strong DNA-binding motif, and in most (55%) cases the known motif was most enriched (P. Kheradpour and M. Kellis, manuscript in preparation).

Protein-binding regions lacking high or moderate affinity cognate recognition sites have 21% lower median scores by rank than regions with recognition sequences (Wilcoxon rank sum P value $<10^{-16}$). Eighty-two per cent of the low-signal regions have high-affinity recognition sequences for other factors. In addition, when ChIP-seq peaks are ranked by their concordance with their known recognition sequence, the median DNase I accessibility is twofold higher in the bottom 20% of peaks than in the upper 80% (genome structure correction (GSC)[20] P value $<10^{-16}$), consistent with previous observations[21–24]. We speculate that low signal regions are either lower-affinity sites[21] or indirect transcription-factor target regions associated through interactions with other factors (see also refs 25, 26).

We organized all the information associated with each transcription factor—including the ChIP-seq peaks, discovered motifs and associated histone modification patterns—in FactorBook (http://www.factorbook.org; ref. 26), a public resource that will be updated as the project proceeds.

DNase I hypersensitive sites and footprints

Chromatin accessibility characterized by DNase I hypersensitivity is the hallmark of regulatory DNA regions[27,28]. We mapped 2.89 million unique, non-overlapping DNase I hypersensitive sites (DHSs) by DNase-seq in 125 cell types, the overwhelming majority of which lie distal to TSSs[29]. We also mapped 4.8 million sites across 25 cell types that displayed reduced nucleosomal crosslinking by FAIRE, many of which coincide with DHSs. In addition, we used micrococcal nuclease to map nucleosome occupancy in GM12878 and K562 cells[30].

In tier 1 and tier 2 cell types, we identified a mean of 205,109 DHSs per cell type (at false discovery rate (FDR) 1%), encompassing an average of 1.0% of the genomic sequence in each cell type, and 3.9% in aggregate. On average, 98.5% of the occupancy sites of transcription factors mapped by ENCODE ChIP-seq (and, collectively, 94.4% of all 1.1 million transcription factor ChIP-seq peaks in K562 cells) lie within accessible chromatin defined by DNase I hotspots[29]. However, a small number of factors, most prominently heterochromatin-bound repressive complexes (for example, the TRIM28–SETDB1–ZNF274 complex[31,32] encoded by the *TRIM28*, *SETDB1* and *ZNF274* genes), seem to occupy a significant fraction of nucleosomal sites.

Using genomic DNase I footprinting[33,34] on 41 cell types we identified 8.4 million distinct DNase I footprints (FDR 1%)[25]. Our *de novo* motif discovery on DNase I footprints recovered ~90% of known transcription factor motifs, together with hundreds of novel evolutionarily conserved motifs, many displaying highly cell-selective occupancy patterns similar to major developmental and tissue-specific regulators.

Regions of histone modification

We assayed chromosomal locations for up to 12 histone modifications and variants in 46 cell types, including a complete matrix of eight modifications across tier 1 and tier 2. Because modification states may span multiple nucleosomes, which themselves can vary in position across cell populations, we used a continuous signal measure of histone modifications in downstream analysis, rather than calling regions (M. M. Hoffman *et al.*, manuscript in preparation; see http://code.google.com/p/align2rawsignal/). For the strongest, 'peak-like' histone modifications, we used MACS[35] to characterize enriched sites. Table 2 describes the different histone modifications, their peak characteristics, and a summary of their known roles (reviewed in refs 36–39).

Our data show that global patterns of modification are highly variable across cell types, in accordance with changes in transcriptional activity. Consistent with previous studies[40,41], we find that integration of the different histone modification information can be used systematically to assign functional attributes to genomic regions (see below).

DNA methylation

Methylation of cytosine, usually at CpG dinucleotides, is involved in epigenetic regulation of gene expression. Promoter methylation is typically associated with repression, whereas genic methylation correlates with transcriptional activity[42]. We used reduced representation bisulphite sequencing (RRBS) to profile DNA methylation quantitatively for an average of 1.2 million CpGs in each of 82 cell lines and tissues (8.6% of non-repetitive genomic CpGs), including CpGs in intergenic regions, proximal promoters and intragenic regions (gene bodies)[43], although it should be noted that the RRBS method preferentially targets CpG-rich islands. We found that 96% of CpGs exhibited differential methylation in at least one cell type or tissue

Table 2 | Summary of ENCODE histone modifications and variants

Histone modification or variant	Signal characteristics	Putative functions
H2A.Z	Peak	Histone protein variant (H2A.Z) associated with regulatory elements with dynamic chromatin
H3K4me1	Peak/region	Mark of regulatory elements associated with enhancers and other distal elements, but also enriched downstream of transcription starts
H3K4me2	Peak	Mark of regulatory elements associated with promoters and enhancers
H3K4me3	Peak	Mark of regulatory elements primarily associated with promoters/transcription starts
H3K9ac	Peak	Mark of active regulatory elements with preference for promoters
H3K9me1	Region	Preference for the 5′ end of genes
H3K9me3	Peak/region	Repressive mark associated with constitutive heterochromatin and repetitive elements
H3K27ac	Peak	Mark of active regulatory elements; may distinguish active enhancers and promoters from their inactive counterparts
H3K27me3	Region	Repressive mark established by polycomb complex activity associated with repressive domains and silent developmental genes
H3K36me3	Region	Elongation mark associated with transcribed portions of genes, with preference for 3′ regions after intron 1
H3K79me2	Region	Transcription-associated mark, with preference for 5′ end of genes
H4K20me1	Region	Preference for 5′ end of genes

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assayed (K. Varley *et al.*, personal communication), and levels of DNA methylation correlated with chromatin accessibility. The most variably methylated CpGs are found more often in gene bodies and intergenic regions, rather than in promoters and upstream regulatory regions. In addition, we identified an unexpected correspondence between unmethylated genic CpG islands and binding by P300, a histone acetyltransferase linked to enhancer activity[44].

Because RRBS is a sequence-based assay with single-base resolution, we were able to identify CpGs with allele-specific methylation consistent with genomic imprinting, and determined that these loci exhibit aberrant methylation in cancer cell lines (K. Varley *et al.*, personal communication). Furthermore, we detected reproducible cytosine methylation outside CpG dinucleotides in adult tissues[45], providing further support that this non-canonical methylation event may have important roles in human biology (K. Varley *et al.*, personal communication).

Chromosome-interacting regions

Physical interaction between distinct chromosome regions that can be separated by hundreds of kilobases is thought to be important in the regulation of gene expression[46]. We used two complementary chromosome conformation capture (3C)-based technologies to probe these long-range physical interactions.

A 3C-carbon copy (5C) approach[47,48] provided unbiased detection of long-range interactions with TSSs in a targeted 1% of the genome (the 44 ENCODE pilot regions) in four cell types (GM12878, K562, HeLa-S3 and H1 hESC)[49]. We discovered hundreds of statistically significant long-range interactions in each cell type after accounting for chromatin polymer behaviour and experimental variation. Pairs of interacting loci showed strong correlation between the gene expression level of the TSS and the presence of specific functional element classes such as enhancers. The average number of distal elements interacting with a TSS was 3.9, and the average number of TSSs interacting with a distal element was 2.5, indicating a complex network of interconnected chromatin. Such interwoven long-range architecture was also uncovered genome-wide using chromatin interaction analysis with paired-end tag sequencing (ChIA-PET)[50] applied to identify interactions in chromatin enriched by RNA polymerase II (Pol II) ChIP from five cell types[51]. In K562 cells, we identified 127,417 promoter-centred chromatin interactions using ChIA-PET, 98% of which were intra-chromosomal. Whereas promoter regions of 2,324 genes were involved in 'single-gene' enhancer–promoter interactions, those of 19,813 genes were involved in 'multi-gene' interaction complexes spanning up to several megabases, including promoter–promoter and enhancer–promoter interactions[51].

These analyses portray a complex landscape of long-range gene–element connectivity across ranges of hundreds of kilobases to several megabases, including interactions among unrelated genes (Supplementary Fig. 1, section Y). Furthermore, in the 5C results, 50–60% of long-range interactions occurred in only one of the four cell lines, indicative of a high degree of tissue specificity for gene–element connectivity[49].

Summary of ENCODE-identified elements

Accounting for all these elements, a surprisingly large amount of the human genome, 80.4%, is covered by at least one ENCODE-identified element (detailed in Supplementary Table 1, section Q). The broadest element class represents the different RNA types, covering 62% of the genome (although the majority is inside of introns or near genes). Regions highly enriched for histone modifications form the next largest class (56.1%). Excluding RNA elements and broad histone elements, 44.2% of the genome is covered. Smaller proportions of the genome are occupied by regions of open chromatin (15.2%) or sites of transcription factor binding (8.1%), with 19.4% covered by at least one DHS or transcription factor ChIP-seq peak across all cell lines. Using our most conservative assessment, 8.5% of bases are covered by either a transcription-factor-binding-site motif (4.6%) or a DHS footprint (5.7%). This, however, is still about 4.5-fold higher than the amount of protein-coding exons, and about twofold higher than the estimated amount of pan-mammalian constraint.

Given that the ENCODE project did not assay all cell types, or all transcription factors, and in particular has sampled few specialized or developmentally restricted cell lineages, these proportions must be underestimates of the total amount of functional bases. However, many assays were performed on more than one cell type, allowing assessment of the rate of discovery of new elements. For both DHSs and CTCF-bound sites, the number of new elements initially increases rapidly with a steep gradient for the saturation curve and then slows with increasing number of cell types (Supplementary Figs 1 and 2, section R). With the current data, at the flattest part of the saturation curve each new cell type adds, on average, 9,500 DHS elements (across 106 cell types) and 500 CTCF-binding elements (across 49 cell types), representing 0.45% of the total element number. We modelled saturation for the DHSs and CTCF-binding sites using a Weibull distribution ($r^2 > 0.999$) and predict saturation at approximately 4.1 million (standard error (s.e.) = 108,000) and 185,100 (s.e. = 18,020) sites, respectively, indicating that we have discovered around half of the estimated total DHSs. These estimates represent a lower bound, but reinforce the observation that there is more non-coding functional DNA than either coding sequence or mammalian evolutionarily constrained bases.

The impact of selection on functional elements

From comparative genomic studies, at least 3–8% of bases are under purifying (negative) selection[4–11], indicating that these bases may potentially be functional. We previously found that 60% of mammalian evolutionarily constrained bases were annotated in the ENCODE pilot project, but also observed that many functional elements lacked evidence of constraint[2], a conclusion substantiated by others[52–54]. The diversity and genome-wide occurrence of functional elements now identified provides an unprecedented opportunity to examine further the forces of negative selection on human functional sequences.

We examined negative selection using two measures that highlight different periods of selection in the human genome. The first measure, inter-species, pan-mammalian constraint (GERP-based scores; 24 mammals[8]), addresses selection during mammalian evolution. The second measure is intra-species constraint estimated from the numbers of variants discovered in human populations using data from the 1000 Genomes project[55], and covers selection over human evolution. In Fig. 1, we plot both these measures of constraint for different classes of identified functional elements, excluding features overlapping exons and promoters that are known to be constrained. Each graph also shows genomic background levels and measures of coding-gene constraint for comparison. Because we plot human population diversity on an inverted scale, elements that are more constrained by negative selection will tend to lie in the upper and right-hand regions of the plot.

For DNase I elements (Fig. 1b) and bound motifs (Fig. 1c), most sets of elements show enrichment in pan-mammalian constraint and decreased human population diversity, although for some cell types the DNase I sites do not seem overall to be subject to pan-mammalian constraint. Bound transcription factor motifs have a natural control from the set of transcription factor motifs with equal sequence potential for binding but without binding evidence from ChIP-seq experiments—in all cases, the bound motifs show both more mammalian constraint and higher suppression of human diversity.

Consistent with previous findings, we do not observe genome-wide evidence for pan-mammalian selection of novel RNA sequences (Fig. 1d). There are also a large number of elements without mammalian constraint, between 17% and 90% for transcription-factor-binding regions as well as DHSs and FAIRE regions. Previous studies could not determine whether these sequences are either biochemically active, but with little overall impact on the organism, or under lineage-specific selection. By isolating sequences preferentially inserted into

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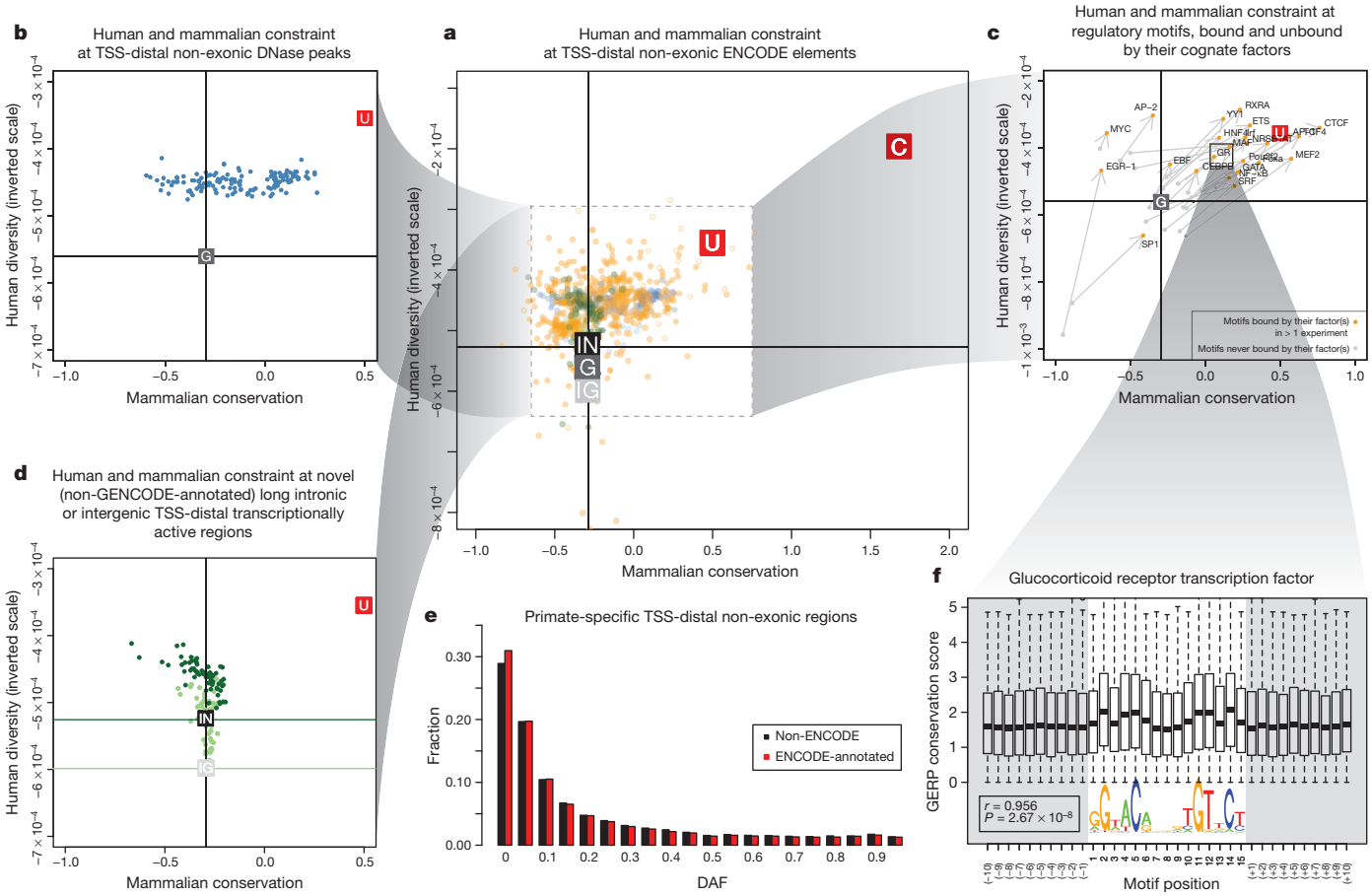

Figure 1 | Impact of selection on ENCODE functional elements in mammals and human populations. **a**, Levels of pan-mammalian constraint (mean GERP score; 24 mammals[8], *x* axis) compared to diversity, a measure of negative selection in the human population (mean expected heterozygosity, inverted scale, *y* axis) for ENCODE data sets. Each point is an average for a single data set. The top-right corners have the strongest evolutionary constraint and lowest diversity. Coding (C), UTR (U), genomic (G), intergenic (IG) and intronic (IN) averages are shown as filled squares. In each case the vertical and horizontal cross hairs show representative levels for the neutral expectation for mammalian conservation and human population diversity, respectively. The spread over all non-exonic ENCODE elements greater than 2.5 kb from TSSs is shown. The inner dashed box indicates that parts of the plot have been magnified for the surrounding outer panels, although the scales in the outer plots provide the exact regions and dimensions magnified. The spread for DHS sites (**b**) and RNA elements (**d**) is shown in the plots on the left. RNA elements are either long novel intronic (dark green) or long intergenic (light green) RNAs. The horizontal cross hairs are colour-coded to the relevant data set in **d**. **c**, Spread of transcription factor motif instances either in regions bound by the transcription factor (orange points) or in the corresponding unbound motif matches in grey, with bound and unbound points connected with an arrow in each case showing that bound sites are generally more constrained and less diverse. **e**, Derived allele frequency spectrum for primate-specific elements, with variations outside ENCODE elements in black and variations covered by ENCODE elements in red. The increase in low-frequency alleles compared to background is indicative of negative selection occurring in the set of variants annotated by the ENCODE data. **f**, Aggregation of mammalian constraint scores over the glucocorticoid receptor (GR) transcription factor motif in bound sites, showing the expected correlation with the information content of bases in the motif. An interactive version of this figure is available in the online version of the paper.

the primate lineage, which is only feasible given the genome-wide scale of this data, we are able to examine this issue specifically. Most primate-specific sequence is due to retrotransposon activity, but an appreciable proportion is non-repetitive primate-specific sequence. Of 104,343,413 primate-specific bases (excluding repetitive elements), 67,769,372 (65%) are found within ENCODE-identified elements. Examination of 227,688 variants segregating in these primate-specific regions revealed that all classes of elements (RNA and regulatory) show depressed derived allele frequencies, consistent with recent negative selection occurring in at least some of these regions (Fig. 1e). An alternative approach examining sequences that are not clearly under pan-mammalian constraint showed a similar result (L. Ward and M. Kellis, manuscript submitted). This indicates that an appreciable proportion of the unconstrained elements are lineage-specific elements required for organismal function, consistent with long-standing views of recent evolution[56], and the remainder are probably 'neutral' elements[2] that are not currently under selection but may still affect cellular or larger scale phenotypes without an effect on fitness.

The binding patterns of transcription factors are not uniform, and we can correlate both inter- and intra-species measures of negative selection with the overall information content of motif positions. The selection on some motif positions is as high as protein-coding exons (Fig. 1f; L. Ward and M. Kellis, manuscript submitted). These aggregate measures across motifs show that the binding preferences found in the population of sites are also relevant to the per-site behaviour. By developing a per-site metric of population effect on bound motifs, we found that highly constrained bound instances across mammals are able to buffer the impact of individual variation[57].

ENCODE data integration with known genomic features

Promoter-anchored integration

Many of the ENCODE assays directly or indirectly provide information about the action of promoters. Focusing on the TSSs of protein-coding transcripts, we investigated the relationships between different ENCODE assays, in particular testing the hypothesis that RNA expression (output) can be effectively predicted from patterns of

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chromatin modification or transcription factor binding (input). Consistent with previous reports[58], we observe two relatively distinct types of promoter: (1) broad, mainly (C+G)-rich, TATA-less promoters; and (2) narrow, TATA-box-containing promoters. These promoters have distinct patterns of histone modifications, and transcription-factor-binding sites are selectively enriched in each class (Supplementary Fig. 1, section Z).

We developed predictive models to explore the interaction between histone modifications and measures of transcription at promoters, distinguishing between modifications known to be added as a consequence of transcription (such as H3K36me3 and H3K79me2) and other categories of histone marks[59]. In our analyses, the best models had two components: an initial classification component (on/off) and a second quantitative model component. Our models showed that activating acetylation marks (H3K27ac and H3K9ac) are roughly as informative as activating methylation marks (H3K4me3 and H3K4me2) (Fig. 2a). Although repressive marks, such as H3K27me3 or H3K9me3, show negative correlation both individually and in the model, removing these marks produces only a small reduction in model performance. However, for a subset of promoters in each cell line, repressive histone marks (H3K27me3 or H3K9me3) must be used to predict their expression accurately. We also examined the interplay between the H3K79me2 and H3K36me3 marks, both of which mark gene bodies, probably reflecting recruitment of modification enzymes by polymerase isoforms. As described previously, H3K79me2 occurs preferentially at the 5′ ends of gene bodies and H3K36me3 occurs more 3′, and our analyses support the previous model in which the H3K79me2 to H3K36me3 transition occurs at the first 3′ splice site[60].

Few previous studies have attempted to build qualitative or quantitative models of transcription genome-wide from transcription factor levels because of the paucity of documented transcription-factor-binding regions and the lack of coordination around a single cell line. We thus examined the predictive capacity of transcription-factor-binding signals for the expression levels of promoters (Fig. 2b).

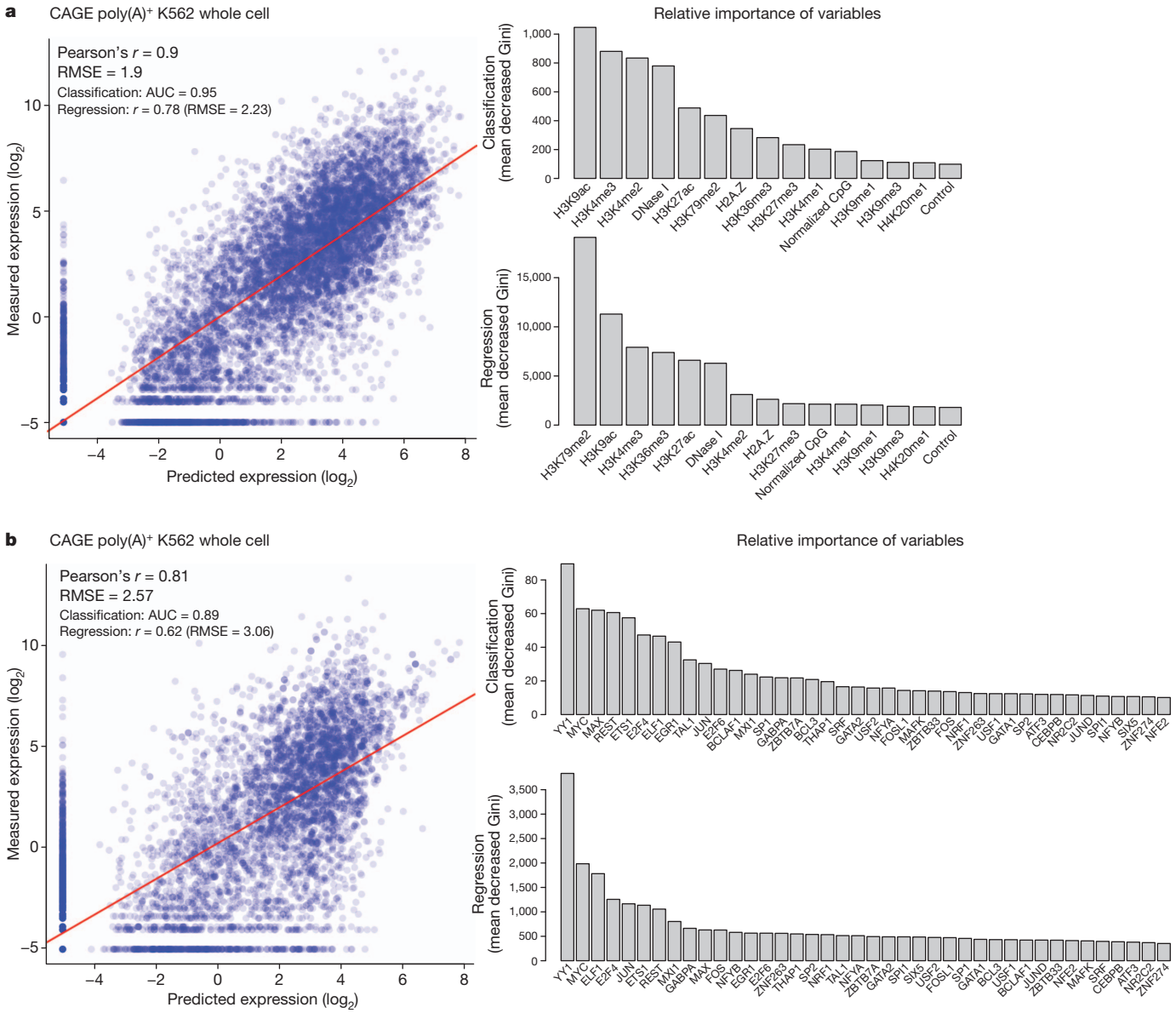

Figure 2 | Modelling transcription levels from histone modification and transcription-factor-binding patterns. **a**, **b**, Correlative models between either histone modifications or transcription factors, respectively, and RNA production as measured by CAGE tag density at TSSs in K562 cells. In each case the scatter plot shows the output of the correlation models (*x* axis) compared to observed values (*y* axis). The bar graphs show the most important histone modifications (**a**) or transcription factors (**b**) in both the initial classification phase (top bar graph) or the quantitative regression phase (bottom bar graph), with larger values indicating increasing importance of the variable in the model. Further analysis of other cell lines and RNA measurement types is reported elsewhere[59,79]. AUC, area under curve; Gini, Gini coefficient; RMSE, root mean square error.

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In contrast to the profiles of histone modifications, most transcription factors show enriched binding signals in a narrow DNA region near the TSS, with relatively higher binding signals in promoters with higher CpG content. Most of this correlation could be recapitulated by looking at the aggregate binding of transcription factors without specific transcription factor terms. Together, these correlation models indicate both that a limited set of chromatin marks are sufficient to 'explain' transcription and that a variety of transcription factors might have broad roles in general transcription levels across many genes. It is important to note that this is an inherently observational study of correlation patterns, and is consistent with a variety of mechanistic models with different causal links between the chromatin, transcription factor and RNA assays. However, it does indicate that there is enough information present at the promoter regions of genes to explain most of the variation in RNA expression.

We developed predictive models similar to those used to model transcriptional activity to explore the relationship between levels of histone modification and inclusion of exons in alternately spliced transcripts. Even accounting for expression level, H3K36me3 has a positive contribution to exon inclusion, whereas H3K79me2 has a negative contribution (H. Tilgner *et al.*, manuscript in preparation). By monitoring the RNA populations in the subcellular fractions of K562 cells, we found that essentially all splicing is co-transcriptional[61], further supporting a link between chromatin structure and splicing.

Transcription-factor-binding site-anchored integration

Transcription-factor-binding sites provide a natural focus around which to explore chromatin properties. Transcription factors are often multifunctional and can bind a variety of genomic loci with different combinations and patterns of chromatin marks and nucleosome organization. Hence, rather than averaging chromatin mark profiles across all binding sites of a transcription factor, we developed a clustering procedure, termed the Clustered Aggregation Tool (CAGT), to identify subsets of binding sites sharing similar but distinct patterns of chromatin mark signal magnitude, shape and hidden directionality[30]. For example, the average profile of the repressive histone mark H3K27me3 over all 55,782 CTCF-binding sites in H1 hESCs shows poor signal enrichment (Fig. 3a). However, after grouping profiles by signal magnitude we found a subset of 9,840 (17.6%) CTCF-binding sites that exhibit significant flanking H3K27me3 signal. Shape and orientation analysis further revealed that the predominant signal profile for H3K27me3 around CTCF peak summits is asymmetric, consistent with a boundary role for some CTCF sites between active and polycomb-silenced domains. Further examples are provided in Supplementary Figs 5 and 6 of section E. For TAF1, predominantly found near TSSs, the asymmetric sites are orientated with the direction of transcription. However, for distal sites, such as those bound by GATA1 and CTCF, we also observed a high proportion of asymmetric histone patterns, although independent of motif directionality. In fact, all transcription-factor-binding data sets in all cell lines show predominantly asymmetric patterns (asymmetry ratio >0.6) for all chromatin marks but not for DNase I signal (Fig. 3b). This indicates that most transcription-factor-bound chromatin events correlate with structured, directional patterns of histone modifications, and that promoter directionality is not the only source of orientation at these sites.

We also examined nucleosome occupancy relative to the symmetry properties of chromatin marks around transcription-factor-binding sites. Around TSSs, there is usually strong asymmetric nucleosome occupancy, often accounting for most of the histone modification signal (for instance, see Supplementary Fig. 4, section E). However, away from TSSs, there is far less concordance. For example, CTCF-binding sites typically show arrays of well-positioned nucleosomes on either side of the peak summit (Supplementary Fig. 1, section E)[62]. Where the flanking chromatin mark signal is high, the signals are often asymmetric, indicating differential marking with histone modifications (Supplementary Figs 2 and 3, section E). Thus, we

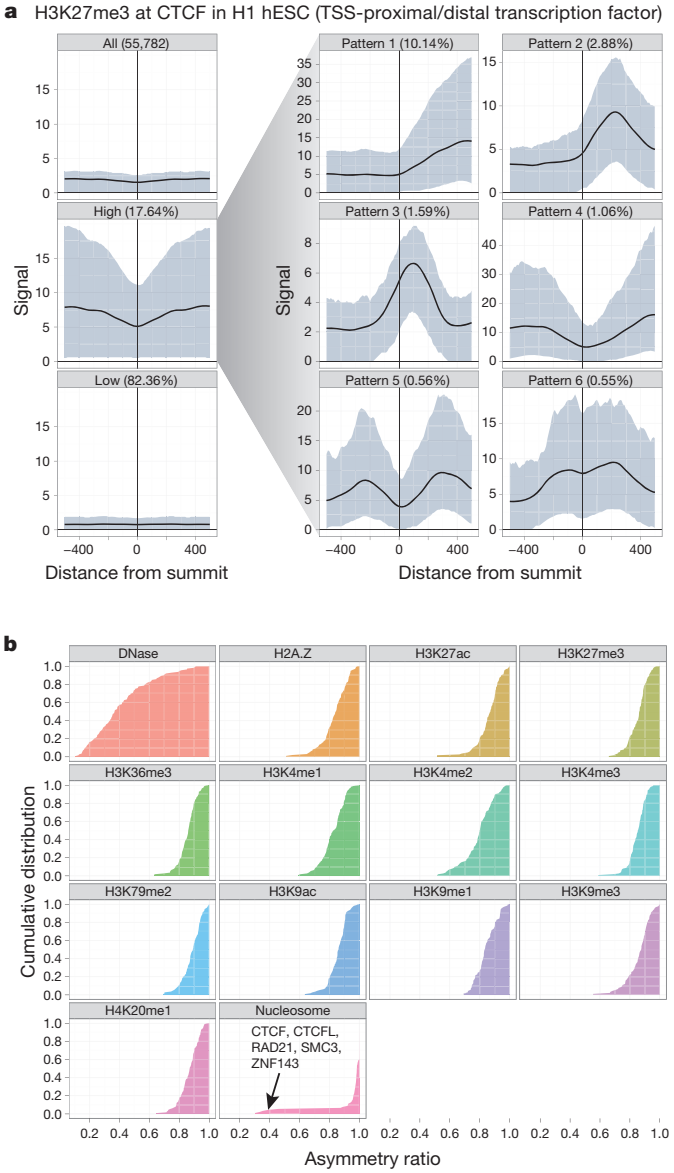

Figure 3 | Patterns and asymmetry of chromatin modification at transcription-factor-binding sites. **a**, Results of clustered aggregation of H3K27me3 modification signal around CTCF-binding sites (a multifunctional protein involved with chromatin structure). The first three plots (left column) show the signal behaviour of the histone modification over all sites (top) and then split into the high and low signal components. The solid lines show the mean signal distribution by relative position with the blue shaded area delimiting the tenth and ninetieth percentile range. The high signal component is then decomposed further into six different shape classes on the right (see ref. 30 for details). The shape decomposition process is strand aware. **b**, Summary of shape asymmetry for DNase I, nucleosome and histone modification signals by plotting an asymmetry ratio for each signal over all transcription-factor-binding sites. All histone modifications measured in this study show predominantly asymmetric patterns at transcription-factor-binding sites. An interactive version of this figure is available in the online version of the paper.

confirm on a genome-wide scale that transcription factors can form barriers around which nucleosomes and histone modifications are arranged in a variety of configurations[62–65]. This is explored in further detail in refs 25, 26 and 30.

Transcription factor co-associations

Transcription-factor-binding regions are nonrandomly distributed across the genome, with respect to both other features (for example, promoters) and other transcription-factor-binding regions. Within the

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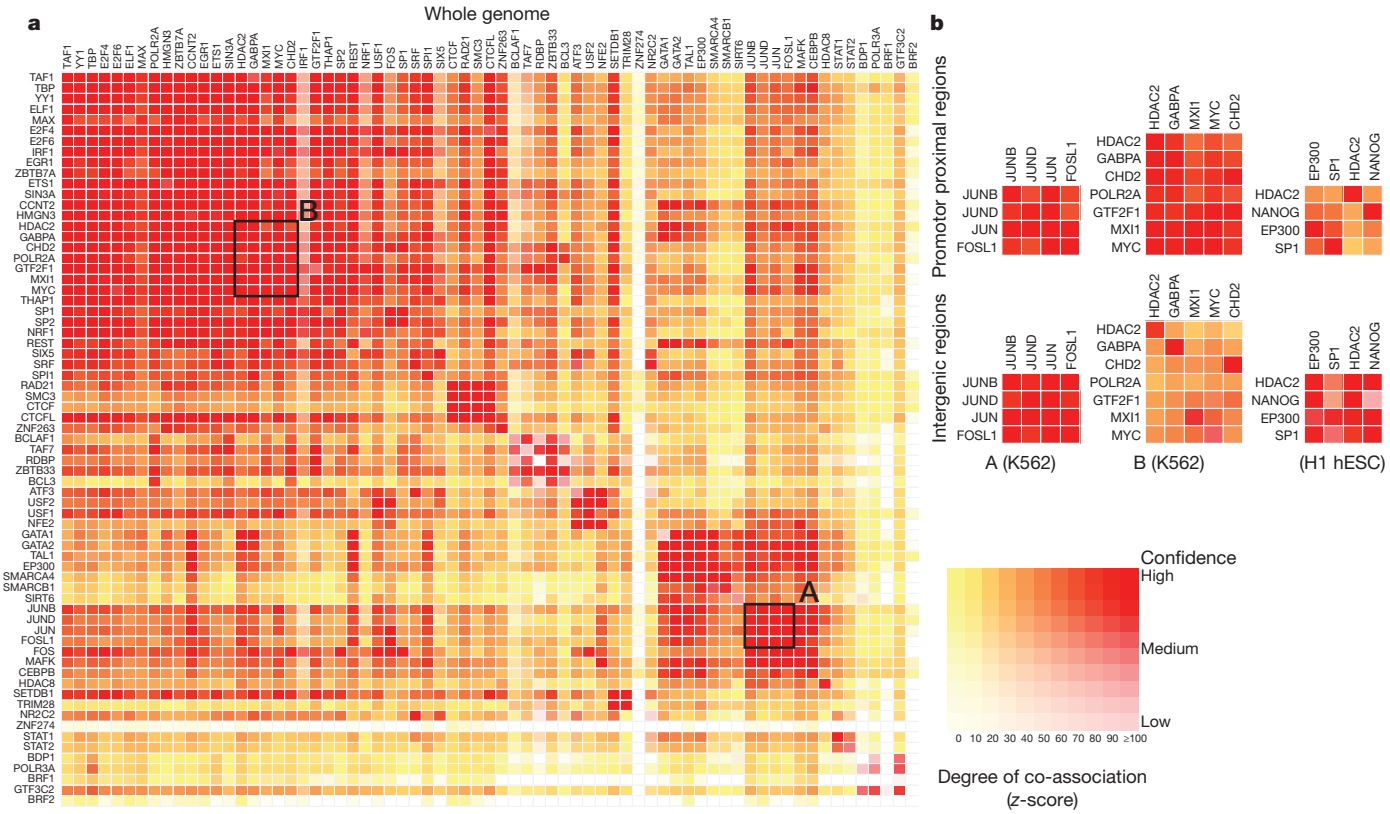

Figure 4 | Co-association between transcription factors. **a**, Significant co-associations of transcription factor pairs using the GSC statistic across the entire genome in K562 cells. The colour strength represents the extent of association (from red (strongest), orange, to yellow (weakest)), whereas the depth of colour represents the fit to the GSC[20] model (where white indicates that the statistical model is not appropriate) as indicated by the key. Most transcription factors have a nonrandom association to other transcription factors, and these associations are dependent on the genomic context, meaning that once the genome is separated into promoter proximal and distal regions, the overall levels of co-association decrease, but more specific relationships are uncovered. **b**, Three classes of behaviour are shown. The first column shows a set of associations for which strength is independent of location in promoter and distal regions, whereas the second column shows a set of transcription factors that have stronger associations in promoter-proximal regions. Both of these examples are from data in K562 cells and are highlighted on the genome-wide co-association matrix (**a**) by the labelled boxes A and B, respectively. The third column shows a set of transcription factors that show stronger association in distal regions (in the H1 hESC line). An interactive version of this figure is available in the online version of the paper.

tier 1 and 2 cell lines, we found 3,307 pairs of statistically co-associated factors ($P < 1 \times 10^{-16}$, GSC) involving 114 out of a possible 117 factors (97%) (Fig. 4a). These include expected associations, such as Jun and Fos, and some less expected novel associations, such as TCF7L2 with HNF4-α and FOXA2 (ref. 66; a full listing is given in Supplementary Table 1, section F). When one considers promoter and intergenic

Table 3 | Summary of the combined state types

Label	Description	Details*	Colour
CTCF	CTCF-enriched element	Sites of CTCF signal lacking histone modifications, often associated with open chromatin. Many probably have a function in insulator assays, but because of the multifunctional nature of CTCF, we are conservative in our description. Also enriched for the cohesin components RAD21 and SMC3; CTCF is known to recruit the cohesin complex.	Turquoise
E	Predicted enhancer	Regions of open chromatin associated with H3K4me1 signal. Enriched for other enhancer-associated marks, including transcription factors known to act at enhancers. In enhancer assays, many of these (>50%) function as enhancers. A more conservative alternative would be *cis*-regulatory regions. Enriched for sites for the proteins encoded by *EP300*, *FOS*, *FOSL1*, *GATA2*, *HDAC8*, *JUNB*, *JUND*, *NFE2*, *SMARCA4*, *SMARCB1*, *SIRT6* and *TAL1* genes in K562 cells. Have nuclear and whole-cell RNA signal, particularly poly(A)− fraction.	Orange
PF	Predicted promoter flanking region	Regions that generally surround TSS segments (see below).	Light red
R	Predicted repressed or low-activity region	This is a merged state that includes H3K27me3 polycomb-enriched regions, along with regions that are silent in terms of observed signal for the input assays to the segmentations (low or no signal). They may have other signals (for example, RNA, not in the segmentation input data). Enriched for sites for the proteins encoded by *REST* and some other factors (for example, proteins encoded by *BRF2*, *CEBPB*, *MAFK*, *TRIM28*, *ZNF274* and *SETDB1* genes in K562 cells).	Grey
TSS	Predicted promoter region including TSS	Found close to or overlapping GENCODE TSS sites. High precision/recall for TSSs. Enriched for H3K4me3. Sites of open chromatin. Enriched for transcription factors known to act close to promoters and polymerases Pol II and Pol III. Short RNAs are most enriched in these segments.	Bright red
T	Predicted transcribed region	Overlap gene bodies with H3K36me3 transcriptional elongation signal. Enriched for phosphorylated form of Pol II signal (elongating polymerase) and poly(A)$^+$ RNA, especially cytoplasmic.	Dark green
WE	Predicted weak enhancer or open chromatin *cis*-regulatory element	Similar to the E state, but weaker signals and weaker enrichments.	Yellow

* Where specific enrichments or overlaps are identified, these are derived from analysis in GM12878 and/or K562 cells where the data for comparison is richest. The colours indicated are used in Figs 5 and 7 and in display of these tracks from the ENCODE data hub.

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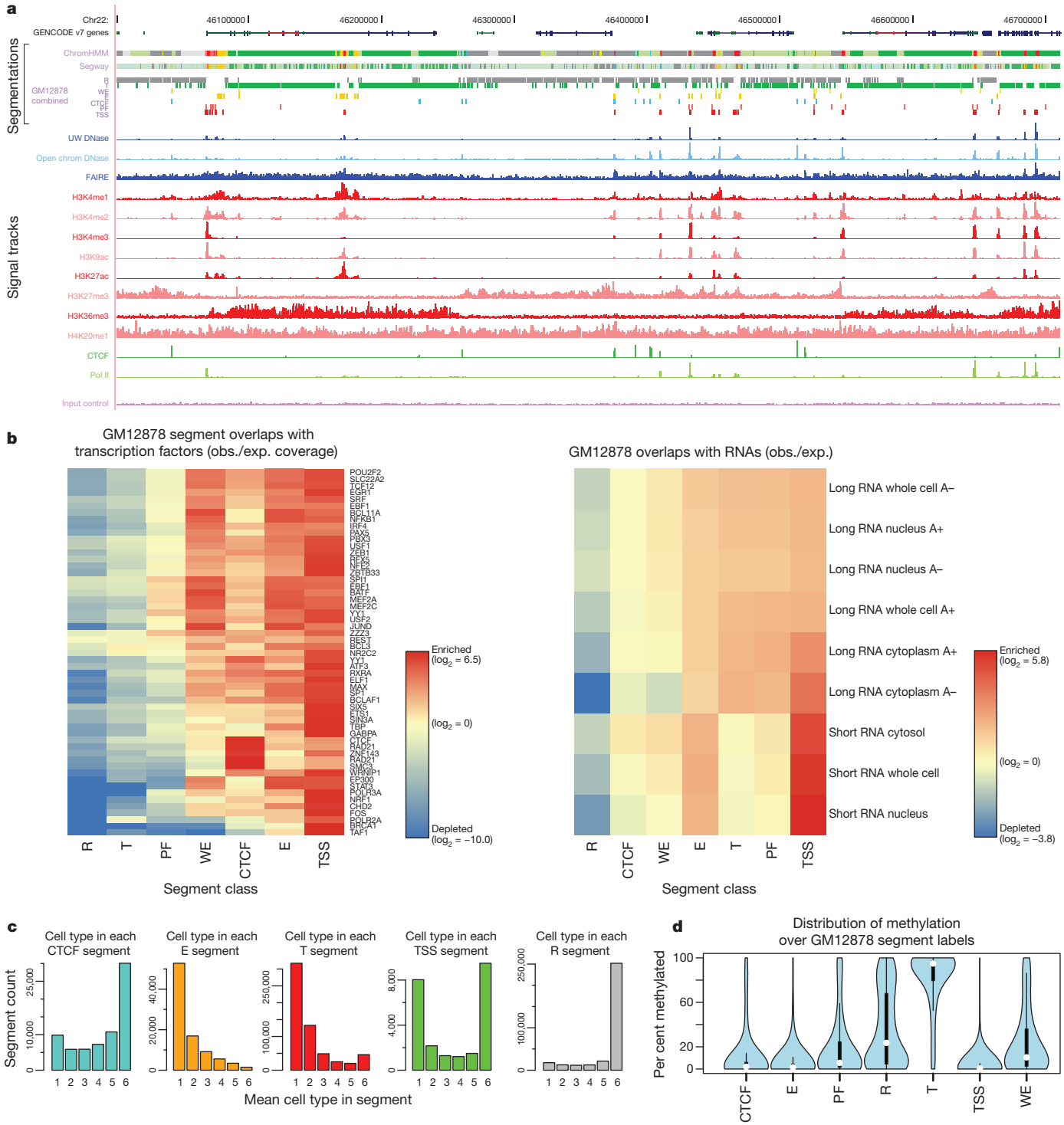

Figure 5 | Integration of ENCODE data by genome-wide segmentation. **a**, Illustrative region with the two segmentation methods (ChromHMM and Segway) in a dense view and the combined segmentation expanded to show each state in GM12878 cells, beneath a compressed view of the GENCODE gene annotations. Note that at this level of zoom and genome browser resolution, some segments appear to overlap although they do not. Segmentation classes are named and coloured according to the scheme in Table 3. Beneath the segmentations are shown each of the normalized signals that were used as the input data for the segmentations. Open chromatin signals from DNase-seq from the University of Washington group (UW DNase) or the ENCODE open chromatin group (Openchrom DNase) and FAIRE assays are shown in blue; signal from histone modification ChIP-seq in red; and transcription factor ChIP-seq signal for Pol II and CTCF in green. The mauve ChIP-seq control signal (input control) at the bottom was also included as an input to the segmentation. **b**, Association of selected transcription factor (left) and RNA (right) elements in the combined segmentation states (*x* axis) expressed as an observed/expected ratio (obs./exp.) for each combination of transcription factor or RNA element and segmentation class using the heat-map scale shown in the key besides each heat map. **c**, Variability of states between cell lines, showing the distribution of occurrences of the state in the six cell lines at specific genome locations: from unique to one cell line to ubiquitous in all six cell lines for five states (CTCF, E, T, TSS and R). **d**, Distribution of methylation level at individual sites from RRBS analysis in GM12878 cells across the different states, showing the expected hypomethylation at TSSs and hypermethylation of genes bodies (T state) and repressed (R) regions.

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regions separately, this changes to 3,201 pairs (116 factors, 99%) for promoters and 1,564 pairs (108 factors, 92%) for intergenic regions, with some associations more specific to these genomic contexts (for example, the cluster of HDAC2, GABPA, CHD2, GTF2F1, MXI1 and MYC in promoter regions and SP1, EP300, HDAC2 and NANOG in intergenic regions (Fig. 4b)). These general and context-dependent associations lead to a network representation of the co-binding with many interesting properties, explored in refs 19, 25 and 26. In addition, we also identified a set of regions bound by multiple factors representing high occupancy of transcription factor (HOT) regions[67].

Genome-wide integration

To identify functional regions genome-wide, we next integrated elements independent of genomic landmarks using either discriminative training methods, where a subset of known elements of a particular class were used to train a model that was then used to discover more instances of this class, or using methods in which only data from ENCODE assays were used without explicit knowledge of any annotation.

For discriminative training, we used a three-step process to predict potential enhancers, described in Supplementary Information and ref. 67. Two alternative discriminative models converged on a set of ~13,000 putative enhancers in K562 cells[67]. In the second approach, two methodologically distinct unbiased approaches (see refs 40, 68 and M. M. Hoffman *et al.*, manuscript in preparation) converged on a concordant set of histone modification and chromatin-accessibility patterns that can be used to segment the genome in each of the tier 1 and tier 2 cell lines, although the individual loci in each state in each cell line are different. With the exception of RNA polymerase II and CTCF, the addition of transcription factor data did not substantially alter these patterns. At this stage, we deliberately excluded RNA and methylation assays, reserving these data as a means to validate the segmentations.

Our integration of the two segmentation methods (M. M. Hoffman *et al.*, manuscript in preparation) established a consensus set of seven major classes of genome states, described in Table 3. The standard view of active promoters, with a distinct core promoter region (TSS and PF states), leading to active gene bodies (T, transcribed state), is rediscovered in this model (Fig. 5a, b). There are three 'active' distal states. We tentatively labelled two as enhancers (predicted enhancers, E, and predicted weak enhancers, WE) due to their occurrence in regions of open chromatin with high H3K4me1, although they differ in the levels of marks such as H3K27ac, currently thought to distinguish active from inactive enhancers. The other active state (CTCF) has high CTCF binding and includes sequences that function as insulators in a transfection assay. The remaining repressed state (R) summarizes sequences split between different classes of actively repressed or inactive, quiescent chromatin. We found that the CTCF-binding-associated state is relatively invariant across cell types, with individual regions frequently occupying the CTCF state across all six cell types (Fig. 5c). Conversely, the E and T states have substantial cell-specific behaviour, whereas the TSS state has a bimodal behaviour with similar numbers of cell-invariant and cell-specific occurrences. It is important to note that the consensus summary classes do not capture all the detail discovered in the individual segmentations containing more states.

The distribution of RNA species across segments is quite distinct, indicating that underlying biological activities are captured in the segmentation. Polyadenylated RNA is heavily enriched in gene bodies. Around promoters, there are short RNA species previously identified as promoter-associated short RNAs (Fig. 5b)[16,69]. Similarly, DNA methylation shows marked distinctions between segments, recapitulating the known biology of predominantly unmethylated active promoters (TSS states) followed by methylated gene bodies[42] (T state, Fig. 5d). The two enhancer-enriched states show distinct patterns of DNA methylation, with the less active enhancer state (by H3K27ac/H3K4me1 levels) showing higher methylation. These states also have an excess of RNA elements without poly(A) tails and methyl-cap RNA, as assayed by CAGE sequences, compared to matched intergenic controls, indicating a specific transcriptional mode associated with active enhancers[70]. Transcription factors also showed distinct distributions across the segments (Fig. 5b). A striking pattern is the concentration of transcription factors in the TSS-associated state. The enhancers contain a different set of transcription factors. For example, in K562 cells, the E state is enriched for binding by the proteins encoded by the *EP300*, *FOS*, *FOSL1*, *GATA2*, *HDAC8*, *JUNB*, *JUND*, *NFE2*, *SMARCA4*, *SMARCB1*, *SIRT6* and *TAL1* genes. We tested a subset of these predicted enhancers in both mouse and fish transgenic models (examples in Fig. 6), with over half of the elements showing activity, often in the corresponding tissue type.

The segmentation provides a linear determination of functional state across the genome, but not an association of particular distal regions with genes. By using the variation of DNase I signal across cell lines, 39% of E (enhancer associated) states could be linked to a proposed regulated gene[29] concordant with physical proximity patterns determined by 5C[49] or ChIA-PET.

To provide a fine-grained regional classification, we turned to a self organizing map (SOM) to cluster genome segmentation regions based on their assay signal characteristics (Fig. 7). The segmentation regions were initially randomly assigned to a 1,350-state map in a two-dimensional toroidal space (Fig. 7a). This map can be visualized as a two-dimensional rectangular plane onto which the various signal distributions can be plotted. For instance, the rectangle at the bottom left of Fig. 7a shows the distribution of the genome in the initial randomized map. The SOM was then trained using the twelve different ChIP-seq and DNase-seq assays in the six cell types previously analysed in the large-scale segmentations (that is, over 72-dimensional space). After training, the SOM clustering was again visualized in two dimensions, now showing the organized distribution of genome segments (lower right of panel, Fig. 7a). Individual data sets associated with the genome segments in each SOM map unit (hexagonal cells) can then be visualized in the same framework to learn how each additional kind of data is distributed on the chromatin state map. Figure 7b shows CAGE/TSS expression data overlaid on the randomly initialized (left) and trained map (right) panels. In this way the trained SOM highlighted cell-type-specific TSS clusters (bottom panels of Fig. 7b), indicating that there are sets of tissue-specific TSSs that are distinguished from each other by subtle combinations of ENCODE

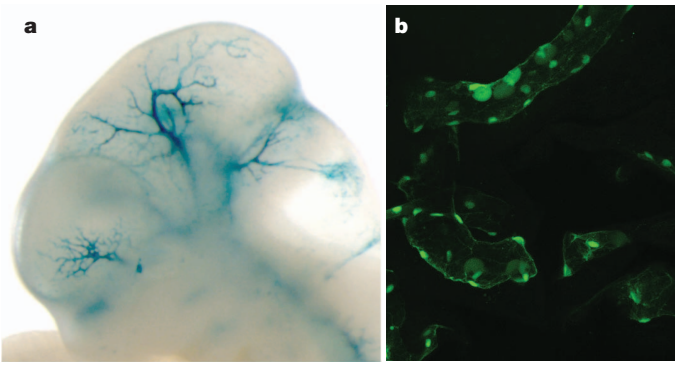

Figure 6 | Experimental characterization of segmentations. Randomly sampled E state segments (see Table 3) from the K562 segmentation were cloned for mouse- and fish-based transgenic enhancer assays. **a**, Representative LacZ-stained transgenic embryonic day (E)11.5 mouse embryo obtained with construct hs2065 (EN167, chr10: 46052882–46055670, GRCh37). Highly reproducible staining in the blood vessels was observed in 9 out of 9 embryos resulting from independent transgenic integration events. **b**, Representative green fluorescent protein reporter transgenic medaka fish obtained from a construct with a basal *hsp70* promoter on meganuclease-based transfection. Reproducible transgenic expression in the circulating nucleated blood cells and the endothelial cell walls was seen in 81 out of 100 transgenic tests of this construct.

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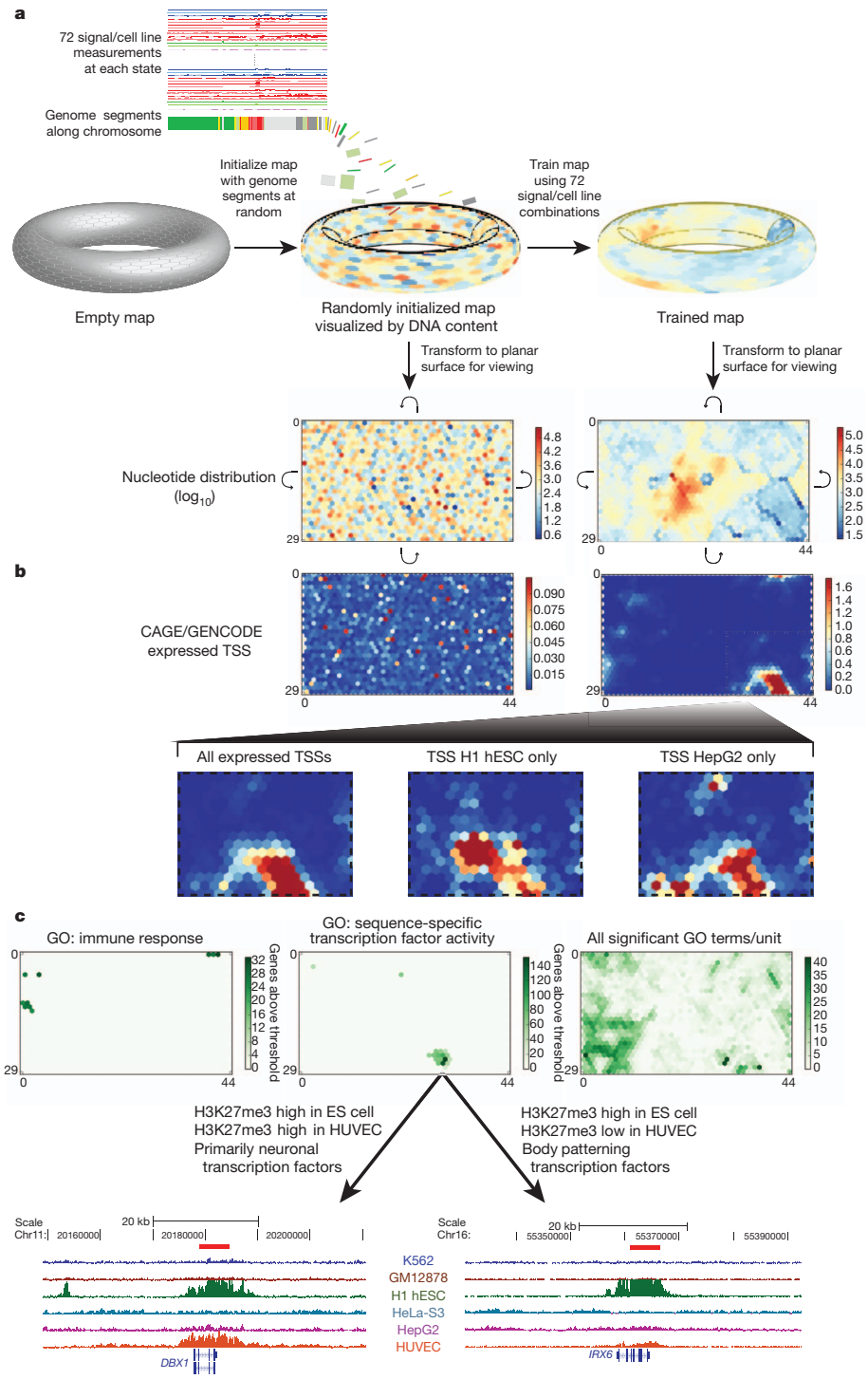

Figure 7 | High-resolution segmentation of ENCODE data by self-organizing maps (SOM). **a–c**, The training of the SOM (**a**) and analysis of the results (**b**, **c**) are shown. Initially we arbitrarily placed genomic segments from the ChromHMM segmentation on to the toroidal map surface, although the SOM does not use the ChromHMM state assignments (**a**). We then trained the map using the signal of the 12 different ChIP-seq and DNase-seq assays in the six cell types analysed. Each unit of the SOM is represented here by a hexagonal cell in a planar two-dimensional view of the toroidal map. Curved arrows indicate that traversing the edges of two dimensional view leads back to the opposite edge. The resulting map can be overlaid with any class of ENCODE or other data to view the distribution of that data within this high-resolution segmentation. In panel **a** the distributions of genome bases across the untrained and trained map (left and right, respectively) are shown using heat-map colours for $\log_{10}$ values. **b**, The distribution of TSSs from CAGE experiments of GENCODE annotation on the planar representations of either the initial random organization (left) or the final trained SOM (right) using heat maps coloured according to the accompanying scales. The bottom half of **b** expands the different distributions in the SOM for all expressed TSSs (left) or TSSs specifically expressed in two example cell lines, H1 hESC (centre) and HepG2 (right). **c**, The association of Gene Ontology (GO) terms on the same representation of the same trained SOM. We assigned genes that are within 20 kb of a genomic segment in a SOM unit to that unit, and then associated this set of genes with GO terms using a hypergeometric distribution after correcting for multiple testing. Map units that are significantly associated to GO terms are coloured green, with increasing strength of colour reflecting increasing numbers of genes significantly associated with the GO terms for either immune response (left) or sequence-specific transcription factor activity (centre). In each case, specific SOM units show association with these terms. The right-hand panel shows the distribution on the same SOM of all significantly associated GO terms, now colouring by GO term count per SOM unit. For sequence-specific transcription factor activity, two example genomic regions are extracted at the bottom of panel **c** from neighbouring SOM units. These are regions around the *DBX1* (from SOM unit 26,31, left panel) and *IRX6* (SOM unit 27,30, right panel) genes, respectively, along with their H3K27me3 ChIP-seq signal for each of the tier 1 and 2 cell types. For *DBX1*, representative of a set of primarily neuronal transcription factors associated with unit 26,31, there is a repressive H3K27me3 signal in both H1 hESCs and HUVECs; for *IRX6*, representative of a set of body patterning transcription factors associated with SOM unit 27,30, the repressive mark is restricted largely to the embryonic stem (ES) cell. An interactive version of this figure is available in the online version of the paper.

chromatin data. Many of the ultra-fine-grained state classifications revealed in the SOM are associated with specific gene ontology (GO) terms (right panel of Fig. 7c). For instance, the left panel of Fig. 7c identifies ten SOM map units enriched with genomic regions associated with genes associated with the GO term 'immune response'. The central panel identifies a different set of map units enriched for the GO term 'sequence-specific transcription factor activity'. The two map units most enriched for this GO term, indicated by the darkest green colouring, contain genes with segments that are high in H3K27me3 in H1 hESCs, but that differ in H3K27me3 levels in HUVECs. Gene function analysis with the GO ontology tool (GREAT[71]) reveals that the map unit with high H3K27me3 levels in both cell types is enriched in transcription factor genes with known neuronal functions, whereas the neighbouring map unit is enriched in genes involved in body patterning. The genome browser shots at the bottom of Fig. 7c pick out an example region for each of the two SOM map units illustrating the difference in H3K27me3 signal. Overall, we have 228 distinct GO terms associated with specific segments across

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one or more states (A. Mortazavi, personal communication), and can assign over one-third of genes to a GO annotation solely on the basis of its multicellular histone patterns. Thus, the SOM analysis provides a fine-grained map of chromatin data across multiple cell types, which can then be used to relate chromatin structure to other data types at differing levels of resolution (for instance, the large cluster of units containing any active TSS, its subclusters composed of units enriched in TSSs active in only one cell type, or individual map units significantly enriched for specific GO terms).

The classifications presented here are necessarily limited by the assays and cell lines studied, and probably contain a number of heterogeneous classes of elements. Nonetheless, robust classifications can be made, allowing a systematic view of the human genome.

Insights into human genomic variation

We next explored the potential impact of sequence variation on ENCODE functional elements. We examined allele-specific variation using results from the GM12878 cells that are derived from an individual (NA12878) sequenced in the 1000 Genomes project, along with her parents. Because ENCODE assays are predominantly sequence-based, the trio design allows each GM12878 data set to be divided by the specific parental contributions at heterozygous sites, producing aggregate haplotypic signals from multiple genomic sites. We examined 193 ENCODE assays for allele-specific biases using 1,409,992 phased, heterozygous SNPs and 167,096 insertions/deletions (indels) (Fig. 8). Alignment biases towards alleles present in the reference genome sequence were avoided using a sequence specifically tailored to the variants and haplotypes present in NA12878 (a 'personalized genome')[72]. We found instances of preferential binding towards each parental allele. For example, comparison of the results from the POLR2A, H3K79me2 and H3K27me3 assays in the region of *NACC2* (Fig. 8a) shows a strong paternal bias for H3K79me2 and POL2RA and a strong maternal bias for H3K27me3, indicating differential activity for the maternal and paternal alleles.

Figure 8b shows the correlation of selected allele-specific signals across the whole genome. For instance, we found a strong allelic correlation between POL2RA and BCLAF1 binding, as well as negative correlation between H3K79me2 and H3K27me3, both at genes (Fig. 8b, below the diagonal, bottom left) and chromosomal segments (top right). Overall, we found that positive allelic correlations among the 193 ENCODE assays are stronger and more frequent than negative correlations. This may be due to preferential capture of accessible alleles and/or the specific histone modification and transcription factor, assays used in the project.

Rare variants, individual genomes and somatic variants

We further investigated the potential functional effects of individual variation in the context of ENCODE annotations. We divided NA12878 variants into common and rare classes, and partitioned these into those overlapping ENCODE annotation (Fig. 9a and Supplementary Tables 1 and 2, section K). We also predicted potential functional effects: for protein-coding genes, these are either non-synonymous SNPs or variants likely to induce loss of function by frame-shift, premature stop, or splice-site disruption; for other regions, these are variants that overlap a transcription-factor-binding site. We found similar numbers of potentially functional variants affecting protein-coding genes or affecting other ENCODE annotations, indicating that many functional variants within individual genomes lie outside exons of protein-coding genes. A more detailed analysis of regulatory variant annotation is described in ref. 73.

To study further the potential effects of NA12878 genome variants on transcription-factor-binding regions, we performed peak calling using a constructed personal diploid genome sequence for NA12878 (ref. 72). We aligned ChIP-seq sequences from GM12878 separately against the maternal and paternal haplotypes. As expected, a greater

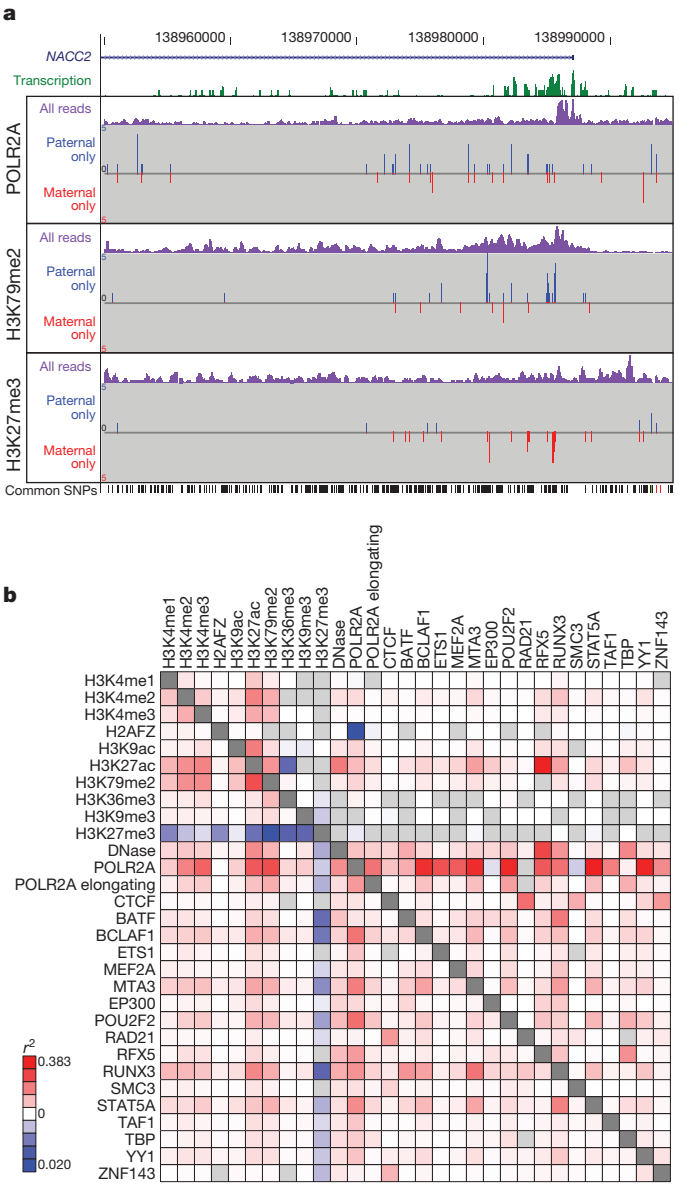

Figure 8 | Allele-specific ENCODE elements. a, Representative allele-specific information from GM12878 cells for selected assays around the first exon of the *NACC2* gene (genomic region Chr9: 138950000–138995000, GRCh37). Transcription signal is shown in green, and the three sections show allele-specific data for three data sets (POLR2A, H3K79me2 and H3K27me3 ChIP-seq). In each case the purple signal is the processed signal for all sequence reads for the assay, whereas the blue and red signals show sequence reads specifically assigned to either the paternal or maternal copies of the genome, respectively. The set of common SNPs from dbSNP, including the phased, heterozygous SNPs used to provide the assignment, are shown at the bottom of the panel. *NACC2* has a statistically significant paternal bias for POLR2A and the transcription-associated mark H3K79me2, and has a significant maternal bias for the repressive mark H3K27me3. **b**, Pair-wise correlations of allele-specific signal within single genes (below the diagonal) or within individual ChromHMM segments across the whole genome for selected DNase-seq and histone modification and transcription factor ChIP-seq assays. The extent of correlation is coloured according to the heat-map scale indicated from positive correlation (red) through to anti-correlation (blue). An interactive version of this figure is available in the online version of the paper.

fraction of reads were aligned than to the reference genome (see Supplementary Information, Supplementary Fig. 1, section K). On average, approximately 1% of transcription-factor-binding sites in GM12878 cells are detected in a haplotype-specific fashion. For instance, Fig. 9b shows a CTCF-binding site not detected using the

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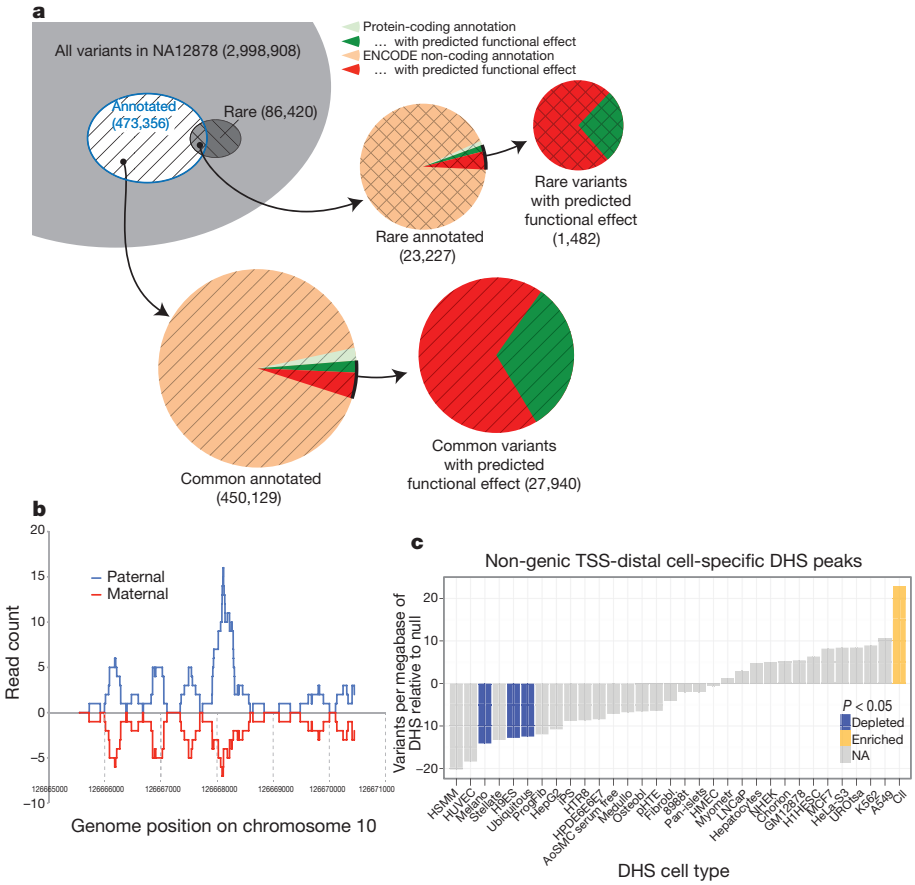

Figure 9 | Examining ENCODE elements on a per individual basis in the normal and cancer genome. **a**, Breakdown of variants in a single genome (NA12878) by both frequency (common or rare (that is, variants not present in the low-coverage sequencing of 179 individuals in the pilot 1 European panel of the 1000 Genomes project[55])) and by ENCODE annotation, including protein-coding gene and non-coding elements (GENCODE annotations for protein-coding genes, pseudogenes and other ncRNAs, as well as transcription-factor-binding sites from ChIP-seq data sets, excluding broad annotations such as histone modifications, segmentations and RNA-seq). Annotation status is further subdivided by predicted functional effect, being non-synonymous and missense mutations for protein-coding regions and variants overlapping bound transcription factor motifs for non-coding element annotations. A substantial proportion of variants are annotated as having predicted functional effects in the non-coding category. **b**, One of several relatively rare occurrences, where alignment to an individual genome sequence (paternal and maternal panels) shows a different readout from the reference genome. In this case, a paternal-haplotype-specific CTCF peak is identified. **c**, Relative level of somatic variants from a whole-genome melanoma sample that occur in DHSs unique to different cell lines. The coloured bars show cases that are significantly enriched or suppressed in somatic mutations. Details of ENCODE cell types can be found at http://encodeproject.org/ENCODE/cellTypes.html. An interactive version of this figure is available in the online version of the paper.

reference sequence that is only present on the paternal haplotype due to a 1-bp deletion (see also Supplementary Fig. 2, section K). As costs of DNA sequencing decrease further, optimized analysis of ENCODE-type data should use the genome sequence of the individual or cell being analysed when possible.

Most analyses of cancer genomes so far have focused on characterizing somatic variants in protein-coding regions. We intersected four available whole-genome cancer data sets with ENCODE annotations (Fig. 9c and Supplementary Fig. 2, section L). Overall, somatic variation is relatively depleted from ENCODE annotated regions, particularly for elements specific to a cell type matching the putative tumour source (for example, skin melanocytes for melanoma). Examining the mutational spectrum of elements in introns for cases where a strand-specific mutation assignment could be made reveals that there are mutational spectrum differences between DHSs and unannotated regions (0.06 Fisher's exact test, Supplementary Fig. 3, section L). The suppression of somatic mutation is consistent with important functional roles of these elements within tumour cells, highlighting a potential alternative set of targets for examination in cancer.

Common variants associated with disease

In recent years, GWAS have greatly extended our knowledge of genetic loci associated with human disease risk and other phenotypes. The output of these studies is a series of SNPs (GWAS SNPs) correlated with a phenotype, although not necessarily the functional variants. Notably, 88% of associated SNPs are either intronic or intergenic[74]. We examined 4,860 SNP–phenotype associations for 4,492 SNPs curated in the National Human Genome Research Institute (NHGRI) GWAS catalogue[74]. We found that 12% of these SNPs overlap transcription-factor-occupied regions whereas 34% overlap DHSs (Fig. 10a). Both figures reflect significant enrichments relative to the overall proportions of 1000 Genomes project SNPs (about 6% and 23%, respectively). Even after accounting for biases introduced by selection of SNPs for the standard genotyping arrays, GWAS SNPs show consistently higher overlap with ENCODE annotations (Fig. 10a, see Supplementary Information). Furthermore, after partitioning the genome by density of different classes of functional elements, GWAS SNPs were consistently enriched beyond all the genotyping SNPs in function-rich partitions, and depleted in function-poor partitions (see Supplementary Fig. 1, section M). GWAS SNPs are particularly enriched in the segmentation classes associated with enhancers and TSSs across several cell types (see Supplementary Fig. 2, section M).

Examining the SOM of integrated ENCODE annotations (see above), we found 19 SOM map units showing significant enrichment for GWAS SNPs, including many SOM units previously associated with specific gene functions, such as the immune response regions.

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Thus, an appreciable proportion of SNPs identified in initial GWAS scans are either functional or lie within the length of an ENCODE annotation (~500 bp on average) and represent plausible candidates for the functional variant. Expanding the set of feasible functional SNPs to those in reasonable linkage disequilibrium, up to 71% of GWAS SNPs have a potential causative SNP overlapping a DNase I site, and 31% of loci have a candidate SNP that overlaps a binding site occupied by a transcription factor (see also refs 73, 75).

The GWAS catalogue provides a rich functional categorization from the precise phenotypes being studied. These phenotypic categorizations are nonrandomly associated with ENCODE annotations and there is marked correspondence between the phenotype and the identity of the cell type or transcription factor used in the ENCODE assay (Fig. 10b). For example, five SNPs associated with Crohn's disease overlap GATA2-binding sites (*P* value 0.003 by random permutation or 0.001 by an empirical approach comparing to the GWAS-matched SNPs; see Supplementary Information), and fourteen are located in DHSs found in immunologically relevant cell types. A notable example is a gene desert on chromosome 5p13.1 containing eight SNPs associated with inflammatory diseases. Several are close to or within DHSs in T-helper type 1 (T_H1) and T_H2 cells as well as peaks of binding by transcription factors in HUVECs (Fig. 10c). The latter cell line is not immunological, but factor occupancy detected there could be a proxy for binding of a more relevant factor, such as GATA3, in T cells. Genetic variants in this region also affect expression levels of *PTGER4* (ref. 76), encoding the prostaglandin receptor EP4. Thus, the ENCODE data reinforce the hypothesis that genetic variants in 5p13.1 modulate the expression of flanking genes, and furthermore provide the specific hypothesis that the variants affect occupancy of a GATA factor in an allele-specific manner, thereby influencing susceptibility to Crohn's disease.

Nonrandom association of phenotypes with ENCODE cell types strengthens the argument that at least some of the GWAS lead SNPs are functional or extremely close to functional variants. Each of the associations between a lead SNP and an ENCODE annotation remains a credible hypothesis of a particular functional element

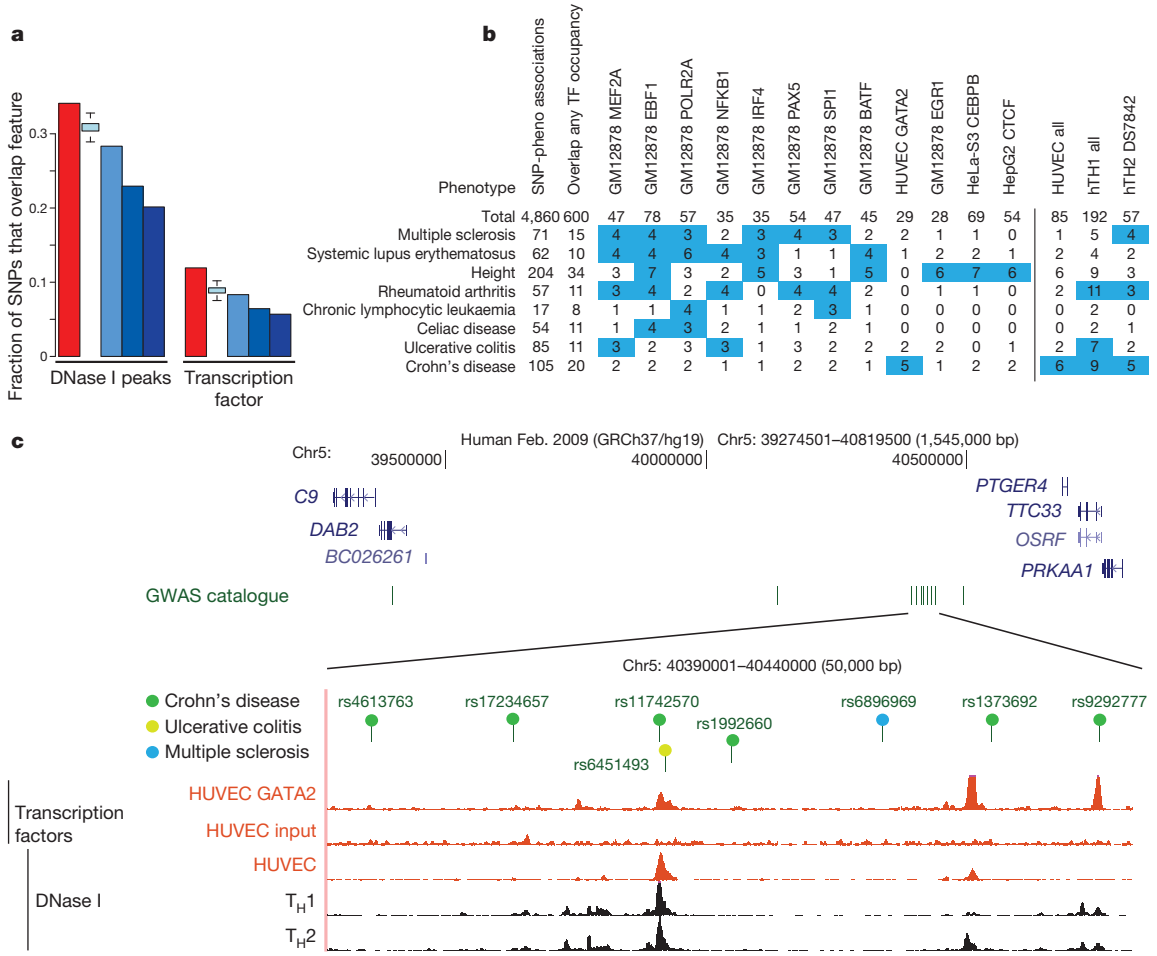

Figure 10 | Comparison of genome-wide-association-study-identified loci with ENCODE data. **a**, Overlap of lead SNPs in the NHGRI GWAS SNP catalogue (June 2011) with DHSs (left) or transcription-factor-binding sites (right) as red bars compared with various control SNP sets in blue. The control SNP sets are (from left to right): SNPs on the Illumina 2.5M chip as an example of a widely used GWAS SNP typing panel; SNPs from the 1000 Genomes project; SNPs extracted from 24 personal genomes (see personal genome variants track at http://main.genome-browser.bx.psu.edu (ref. 80)), all shown as blue bars. In addition, a further control used 1,000 randomizations from the genotyping SNP panel, matching the SNPs with each NHGRI catalogue SNP for allele frequency and distance to the nearest TSS (light blue bars with bounds at 1.5 times the interquartile range). For both DHSs and transcription-factor-binding regions, a larger proportion of overlaps with GWAS-implicated SNPs is found compared to any of the controls sets. **b**, Aggregate overlap of phenotypes to selected transcription-factor-binding sites (left matrix) or DHSs in selected cell lines (right matrix), with a count of overlaps between the phenotype and the cell line/factor. Values in blue squares pass an empirical *P*-value threshold ≤0.01 (based on the same analysis of overlaps between randomly chosen, GWAS-matched SNPs and these epigenetic features) and have at least a count of three overlaps. The *P* value for the total number of phenotype–transcription factor associations is <0.001. **c**, Several SNPs associated with Crohn's disease and other inflammatory diseases that reside in a large gene desert on chromosome 5, along with some epigenetic features indicative of function. The SNP (rs11742570) strongly associated to Crohn's disease overlaps a GATA2 transcription-factor-binding signal determined in HUVECs. This region is also DNase I hypersensitive in HUVECs and T-helper T_H1 and T_H2 cells. An interactive version of this figure is available in the online version of the paper.

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class or cell type to explore with future experiments. Supplementary Tables 1–3, section M, list all 14,885 pairwise associations across the ENCODE annotations. The accompanying papers have a more detailed examination of common variants with other regulatory information[19,25,29,73,75,77].

Concluding remarks

The unprecedented number of functional elements identified in this study provides a valuable resource to the scientific community as well as significantly enhances our understanding of the human genome. Our analyses have revealed many novel aspects of gene expression and regulation as well as the organization of such information, as illustrated by the accompanying papers (see http://www.encodeproject.org/ENCODE/pubs.html for collected ENCODE publications). However, there are still many specific details, particularly about the mechanistic processes that generate these elements and how and where they function, that require additional experiments to elucidate.

The large spread of coverage—from our highest resolution, most conservative set of bases implicated in GENCODE protein-coding gene exons (2.9%) or specific protein DNA binding (8.5%) to the broadest, most general set of marks covering the genome (approximately 80%), with many gradations in between—presents a spectrum of elements with different functional properties discovered by ENCODE. A total of 99% of the known bases in the genome are within 1.7 kb of any ENCODE element, whereas 95% of bases are within 8 kb of a bound transcription factor motif or DNase I footprint. Interestingly, even using the most conservative estimates, the fraction of bases likely to be involved in direct gene regulation, even though incomplete, is significantly higher than that ascribed to protein-coding exons (1.2%), raising the possibility that more information in the human genome may be important for gene regulation than for biochemical function. Many of the regulatory elements are not constrained across mammalian evolution, which so far has been one of the most reliable indications of an important biochemical event for the organism. Thus, our data provide orthologous indicators for suggesting possible functional elements.

Importantly, for the first time we have sufficient statistical power to assess the impact of negative selection on primate-specific elements, and all ENCODE classes display evidence of negative selection in these unique-to-primate elements. Furthermore, even with our most conservative estimate of functional elements (8.5% of putative DNA/protein binding regions) and assuming that we have already sampled half of the elements from our transcription factor and cell-type diversity, one would estimate that at a minimum 20% (17% from protein binding and 2.9% protein coding gene exons) of the genome participates in these specific functions, with the likely figure significantly higher.

The broad coverage of ENCODE annotations enhances our understanding of common diseases with a genetic component, rare genetic diseases, and cancer, as shown by our ability to link otherwise anonymous associations to a functional element. ENCODE and similar studies provide a first step towards interpreting the rest of the genome—beyond protein-coding genes—thereby augmenting common disease genetic studies with testable hypotheses. Such information justifies performing whole-genome sequencing (rather than exome only, 1.2% of the genome) on rare diseases and investigating somatic variants in non-coding functional elements, for instance, in cancer. Furthermore, as GWAS analyses typically associate disease to SNPs in large regions, comparison to ENCODE non-coding functional elements can help pinpoint putative causal variants in addition to refinement of location by fine-mapping techniques[78]. Combining ENCODE data with allele-specific information derived from individual genome sequences provides specific insight on the impact of a genetic variant. Indeed, we believe that a significant goal would be to use functional data such as that derived from this project to assign every genomic variant to its possible impact on human phenotypes.

So far, ENCODE has sampled 119 of 1,800 known transcription factors and general components of the transcriptional machinery on a limited number of cell types, and 13 of more than 60 currently known histone or DNA modifications across 147 cell types. DNase I, FAIRE and extensive RNA assays across subcellular fractionations have been undertaken on many cell types, but overall these data reflect a minor fraction of the potential functional information encoded in the human genome. An important future goal will be to enlarge this data set to additional factors, modifications and cell types, complementing the other related projects in this area (for example, Roadmap Epigenomics Project, http://www.roadmapepigenomics.org/, and International Human Epigenome Consortium, http://www.ihec-epigenomes.org/). These projects will constitute foundational resources for human genomics, allowing a deeper interpretation of the organization of gene and regulatory information and the mechanisms of regulation, and thereby provide important insights into human health and disease. Co-published ENCODE-related papers can be explored online via the *Nature* ENCODE explorer (http://www.nature.com/ENCODE), a specially designed visualization tool that allows users to access the linked papers and investigate topics that are discussed in multiple papers via thematically organized threads.

METHODS SUMMARY

For full details of Methods, see Supplementary Information.

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Author Contributions See the consortium author list for details of author contributions.

Author Information The Supplementary Information is accompanied by a Virtual Machine (VM) containing the functioning analysis data and code. Further details of the VM are available from http://encodeproject.org/ENCODE/integrativeAnalysis/VM. Reprints and permissions information is available at www.nature.com/reprints. This paper is distributed under the terms of the Creative Commons Attribution-Non-Commercial-Share Alike licence, and the online version of the paper is freely available to all readers. The authors declare no competing financial interests. Readers are welcome to comment on the online version of the paper. Correspondence and requests for materials should be addressed to E.B. (birney@ebi.ac.uk).

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